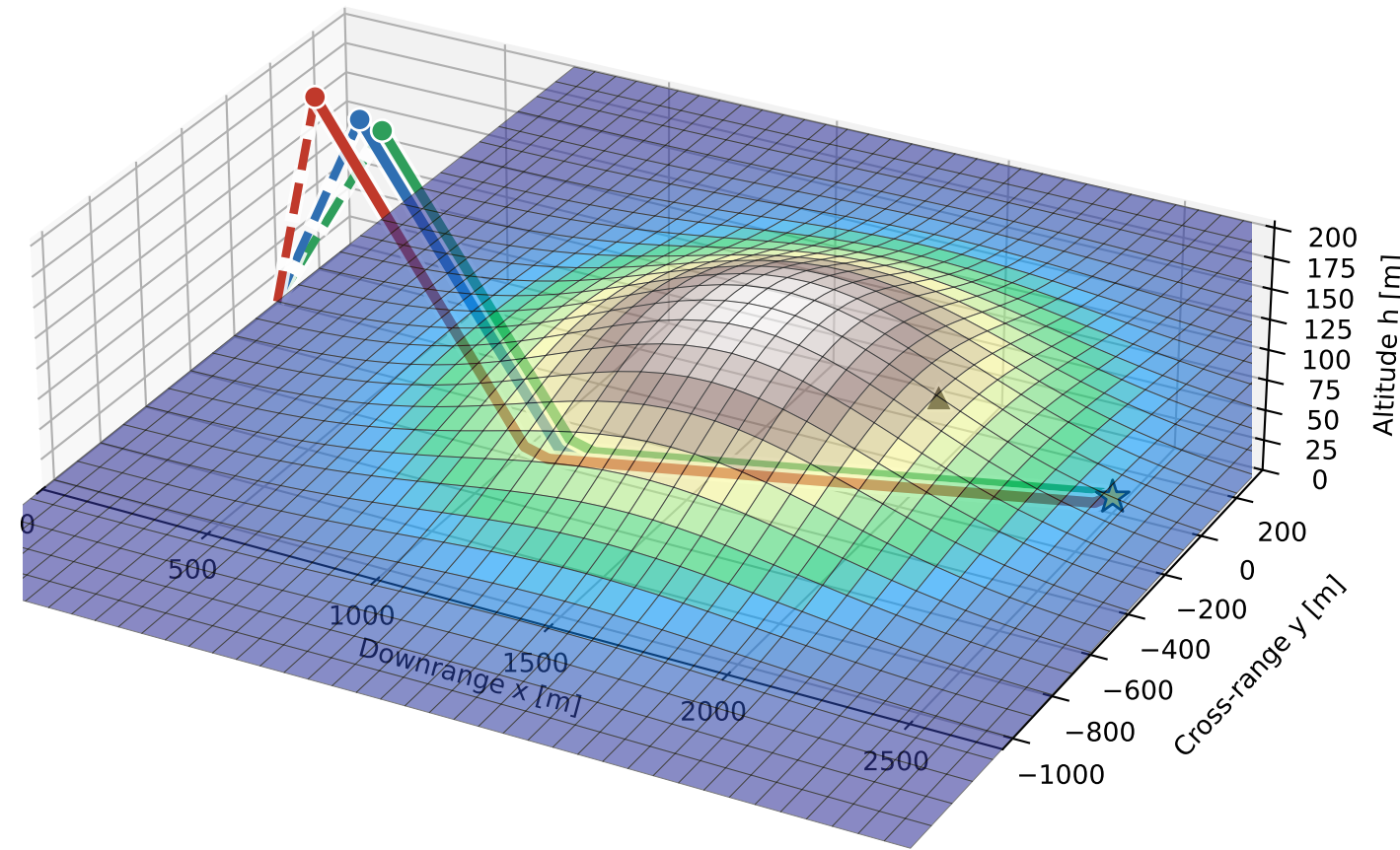


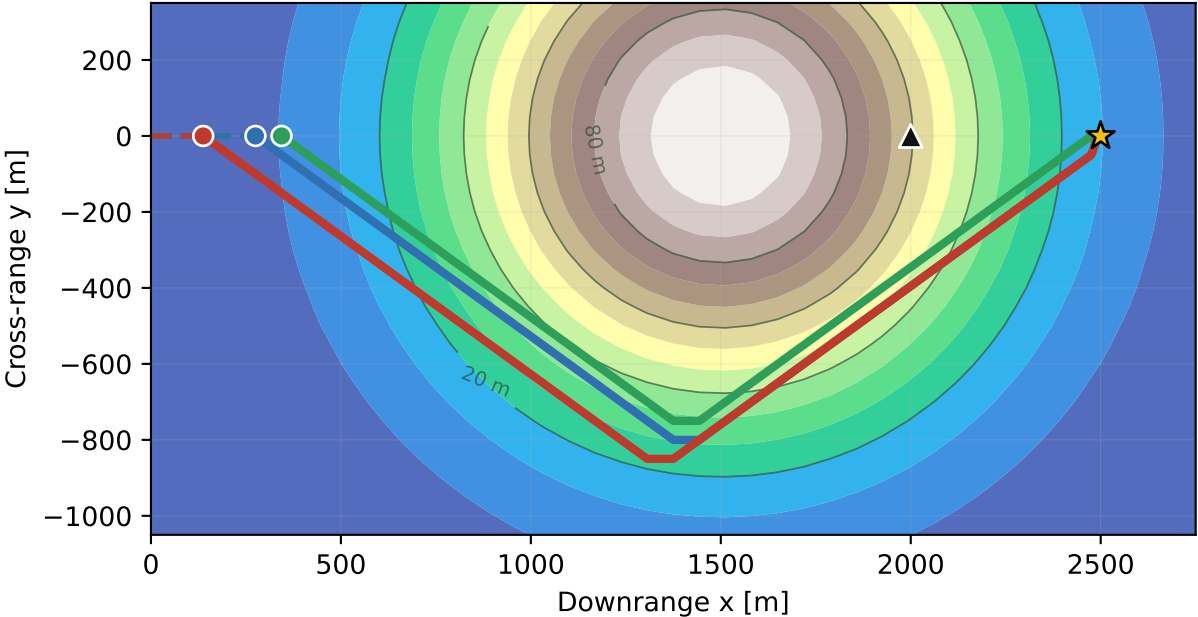
3D Extension: Detection-Time Weighting Changes the Lateral Route

Same terrain and sensor; only the attacker objective weights change — coarse-grid qualitative prototype

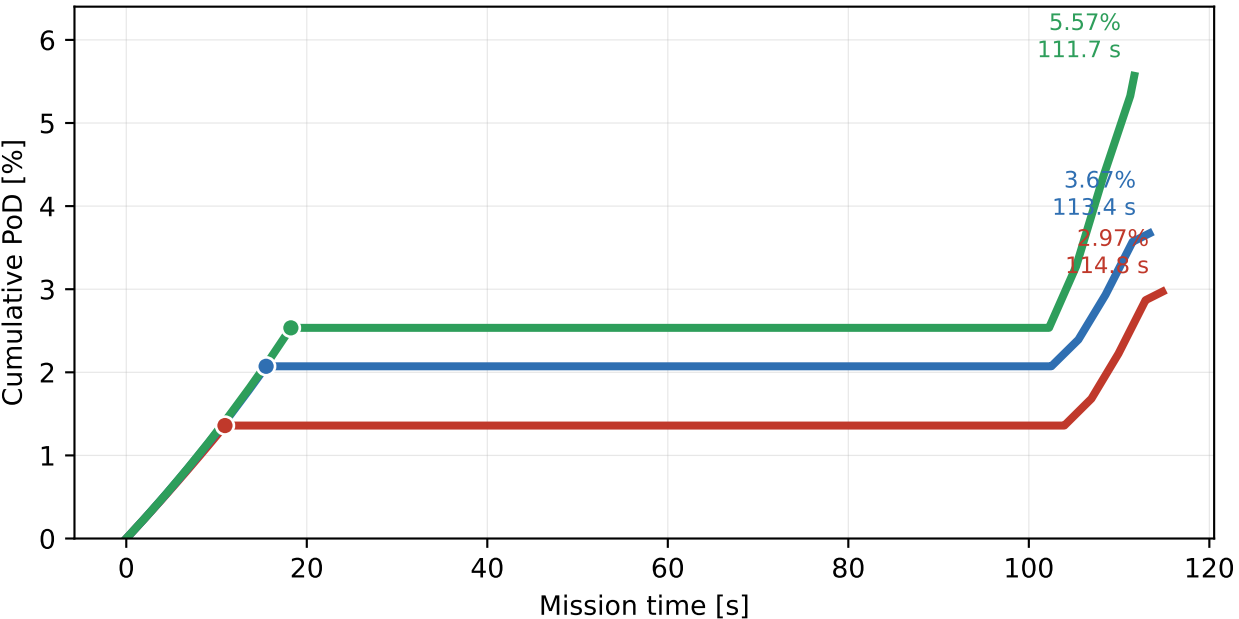
3D route comparison



Top-down lateral route choice



Detection-time trade-off



Detection priority (wPoD=0.75, y_min=-850 m, PoD=2.97%, T=114.8 s) Time priority (wPoD=0.25, y_min=-750 m, PoD=5.57%, T=111.7 s) Glide phase Goal
Balanced (wPoD=0.50, y_min=-800 m, PoD=3.67%, T=113.4 s) Powered phase Sensor